

Java: Variables & Types

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1. Variables

A variable is a named memory location that holds a data value. Key points:

- Declared once; different types use different amounts of memory (compiler reserves it).
- Values can be assigned and changed later.
- Operations are only valid between **compatible types**.

Example:

```
int width = 3;    int height = 4;    int area = width * height;    // area = 12
```

2. Variable Names (Identifiers)

Rules (applies to all Java identifiers):

- Must start with a letter (a–z, A–Z), dollar sign (\$), or underscore (_).
- Followed by letters, digits (0–9), \$, or _.
- Case-sensitive: total ≠ Total ≠ TOTAL.
- Cannot be a reserved keyword (int, if, class, void, public, private, ...).

Good naming guidelines:

- Avoid \$ (reserved for system names).
- Don't rely on case differences alone (total vs Total).
- Use meaningful, moderately-length names — e.g. **currentItem** (not crltn or theCurrentItemBeingProcessed).
- Use **camelCase**: variables/methods start lowercase (dataList2), classes start uppercase (MyFavoriteClass).

Valid vs Invalid examples:

Valid	Invalid (reason)
R2D2, INT, _dogma_95_	30DayAbs — starts with digit
riteOnThru	pork&beans — '&' illegal
SchultzieVonWienerschnitzelIII	private — reserved name
	C-3PO — '-' illegal

3. Primitive Data Types

Java is a **strongly-typed** language — every expression must use compatible types.

Category	Type	Size	Range / Notes
Integer	byte	1 byte	-128 to +127

Integer	short	2 bytes	~-32,000 to +32,000
Integer	int	4 bytes	~-2 billion to +2 billion
Integer	long	8 bytes	Very large range; append L
Float	float	4 bytes	~7 digits precision; append F
Float	double	8 bytes	~15 digits precision (default)
Other	boolean	1 byte	true or false
Other	char	2 bytes	Single Unicode character

Note: String is NOT a primitive — it is an object.

4. Numeric Constants (Literals)

- Integer constants are **int** by default; append **L** for long (e.g. 13943L).
- Floating-point constants are **double** by default; append **F** for float (e.g. 3.14F).
- Doubles accept decimal (3.14159) or scientific notation (3.145E5 = 314500.0).

5. Character & String Constants

char — single character in single quotes: 'A', '9', '*'

String — zero or more characters in double quotes: "Hello"

Escape sequences:

Sequence	Meaning
\\	Backslash
\'	Single quote
\"	Double quote
\n	New line
\t	Tab

6. Operators

Arithmetic:

- Unary negation: **-x**
- Multiply / Divide: **x * y** **x / y** — integer division truncates (23/4 → 5)
- Modulo (remainder): **x % y** (23%4 → 3)
- Real division: **23.0/4.0 → 5.75**
- Add / Subtract: **x+y** **x-y**

Comparison (all return boolean):

x==y **x!=y** **x<y** **x>y** **x<=y** **x>=y**

7. String Operators

- **Concatenation (+)**: joins strings — "von"+"Wiener" → "vonWiener" (no space added). Non-strings are converted first: (8*4)+"degrees" → "32degrees".
- **s.equals(t)** — returns true if strings are equal (use instead of ==).
- **s.length()** — returns number of characters.
- **s.compareTo(t)** — lexicographic comparison: <0 if s<t, 0 if equal, >0 if s>t.